

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'A packet filter firewall filters at the' is 'Network layer'.
2. True or False: The correct answer to '_____ is also a part of darknet that is employed for transferring files anonymously.' is 'Tor'.
3. True or False: The correct answer to 'Which of the following is an encryption method?' is 'All of the above'.
4. True or False: The correct answer to 'A _____ is a sequential segment of the memory location that is allocated for containing some data such as a character string or an...' is 'array'.
5. True or False: The correct answer to 'The _____ is anything which your search engine cannot search.' is 'Deep web'.
6. True or False: The correct answer to 'Why apps developed in languages like C, C++ is prone to Buffer-overflow?' is 'Efficient memory usage'.
7. True or False: The correct answer to 'How does a buffer overflow on the stack facilitate running attacker-injected code?' is 'By overwriting return address'.
8. True or False: The correct answer to 'A/an _____ is a program that steals your logins & passwords for instant messaging applications.' is 'Keylogger'.
9. True or False: The correct answer to 'A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.' is 'firewall'.
10. True or False: The correct answer to 'Let $I = (S,V)$ be a MAC. Suppose $S(k,m)$ is always 10 bits long. Can this MAC be secure?' is 'It depends on the details of the MAC'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to check whether the given number is palindromic or not.
2. Write a function to filter a list by its integer element.

End of Paper